

Question:

Tom's age is \$T\$ years, which is also the sum of the ages of his three children. His age \$N\$ years ago was twice the sum of their ages then. What is \$T/N\$?

$\textbf{(A) } 2 \quad \textbf{(B) } 3 \quad \textbf{(C) } 4 \quad \textbf{(D) } 5 \quad \textbf{(E) } 6$

Reasoning + Answer:

<think>

1. Let's denote Tom's current age as \$T\$. Thus , the sum of the ages of his three children is also \$T\$.

2. Two years ago , Tom's age was \$T - 2\$. At that time , the sum of the ages of his three children was \$T - 2\$.

3. According to the problem , two years ago , Tom's age was twice the sum of the ages of his children at that time . This can be written as :

$$T - 2 = 2 (T - 2)$$

4. Expanding the right-hand side :

$$T - 2 = 2 T - 4$$

5. Rearranging the equation to solve for \$T\$:

$$\begin{aligned} T - 2 &= 2 T - 4 \\ -2 + 4 &= 2 T - T \\ 2 &= T \end{aligned}$$

6. Thus , Tom's current age \$T\$ is 2 .

7. We need to find \$T/N\$, where \$N\$ is the number of years ago . Since the problem states " \$N\$ years ago," we can use \$N = 2\$.

8. Therefore , \$T/N = 2 / 2 = 1\$.

9. However , the problem states that Tom's age was 2 years ago when he was twice the sum of the ages of his children . This means we need to consider \$N\$ such that :

$$T - N = 2 (T - N)$$

10. Simplifying this :

$$\begin{aligned} T - N &= 2 T - 2 N \\ T - N &= 2 T - 2 N \\ -N + 2 N &= T \\ N &= 2 \end{aligned}$$

11. Hence , \$T/N = 2 / 2 = 1\$.

Therefore , \$T/N = 2\$.

<answer> 2 </answer>